MILITARY STANDARD

INTERFACE DESIGNS AND DIMENSIONS FOR FIBER OPTIC INTERCONNECTION DEVICES

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(Project 6060-0038)

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MIL—STD—1863A
23 JANUARY 1985
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MILITARY STANDARD

INTERFACE DESIGNS AND DIMENSIONS FOR FIBER OPTIC INTERCONNECTION DEVICES

FSC 6060

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10 JUNE 1986

DEPARTMENT OF DEFENSE
Washington, DC 20301

INTERFACE DESIGNS AND DIMENSIONS FOR FIBER OPTIC INTERCONNECTION DEVICES

MIL-STD-1863A

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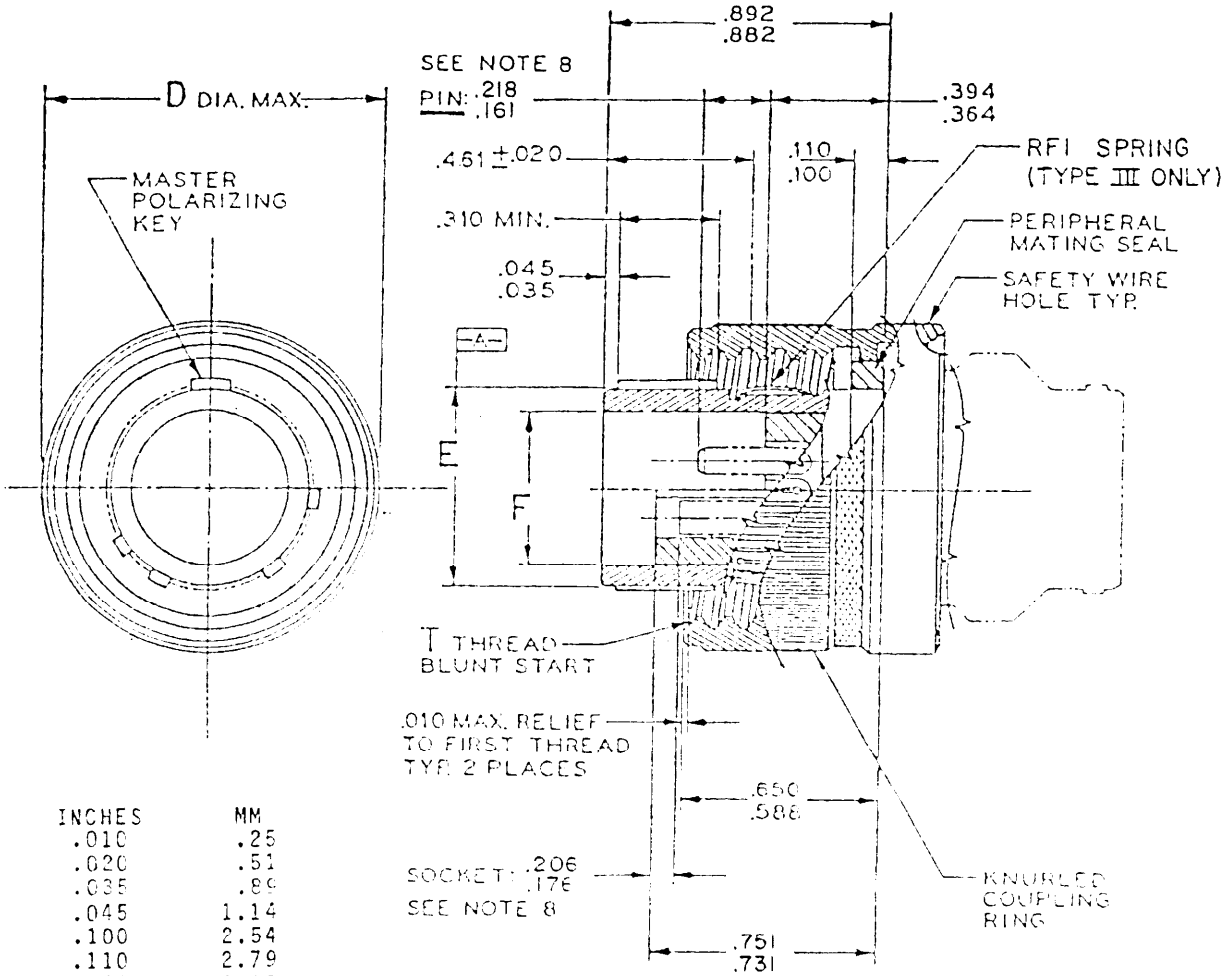
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INCHES	MM
.010	.25
.020	.51
.035	.89
.045	1.14
.100	2.54
.110	2.79
.161	4.09
.176	4.47
.206	5.23
.218	5.54
.310	7.87
.364	9.25
.394	10.01
.461	11.71
.588	14.94
.650	16.51
.731	18.57
.751	19.08
.882	22.40
.892	22.66

FIGURE 1. Interface dimensions for fiber optic connector plug, series 1 - type I, class A and type III, class A, style S.

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Shell sizes	T thread classes 2A	A DIA. MAX JAM-NUT	B DIA.	C DIA
(11)	.750-.1P-.2L-D.S.	1.290 (32.77)	.513 (13.03) .504 (12.80)	.367 (9.32) .352 (8.94)
(13)	.875-.1P-.2L-D.S.	1.435 (36.45)	.637 (16.18) .628 (15.95)	.489 (12.42) .474 (12.04)
(15)	1.062-.1P-.2L-D.S.	1.650 (41.91)	.809 (20.55) .800 (20.32)	.667 (16.94) .652 (16.56)
(17)	1.125-.1P-.2L-D.S.	1.725 (43.82)	.879 (22.33) .870 (22.10)	.744 (18.90) .729 (18.52)
(19)	1.312-.1P-.2L-D.S.	1.940 (49.28)	1.046 (26.57) 1.037 (26.34)	.888 (22.56) .873 (22.17)
(23)	1.500-.1P-.2L-D.S.	2.155 (54.74)	1.231 (31.27) 1.222 (31.04)	1.073 (27.25) 1.058 (26.87)
(25)	1.625-.1P-.2L-D.S.	2.300 (58.42)	1.392 (35.36) 1.383 (35.13)	1.262 (32.05) 1.247 (31.67)
(29)	1.812-.1P-.2L-D.S.	2.515 (63.88)	1.558 (39.57) 1.549 (39.34)	1.412 (35.86) 1.397 (35.48)
(33)	2.000-.1P-.2L-D.S.	2.730 (69.34)	1.750 (44.45) 1.741 (44.22)	1.610 (40.89) 1.595 (40.51)

NOTES:

1. Dimensions are in inches.
2. Millimeters are in parentheses.
3. Metric equivalents (to the nearest .01 mm) are given for general information only.
4. Dimensions apply after plating.
5. Mating keyway positions and dimensions are shown on figure 8.
6. This design information establishes connector intermating criteria and interchangeability of accessory hardware.
7. Rear end connector design for attachment of non-rotatable accessory hardware is shown on figure 11.
8. Dimension includes terminal slack. (Pin front end moves back during mating.)

FIGURE 2. Interface dimensions for fiber optic connector receptacle, series 1 - type I, class A and type III, class A, style S - Continued.

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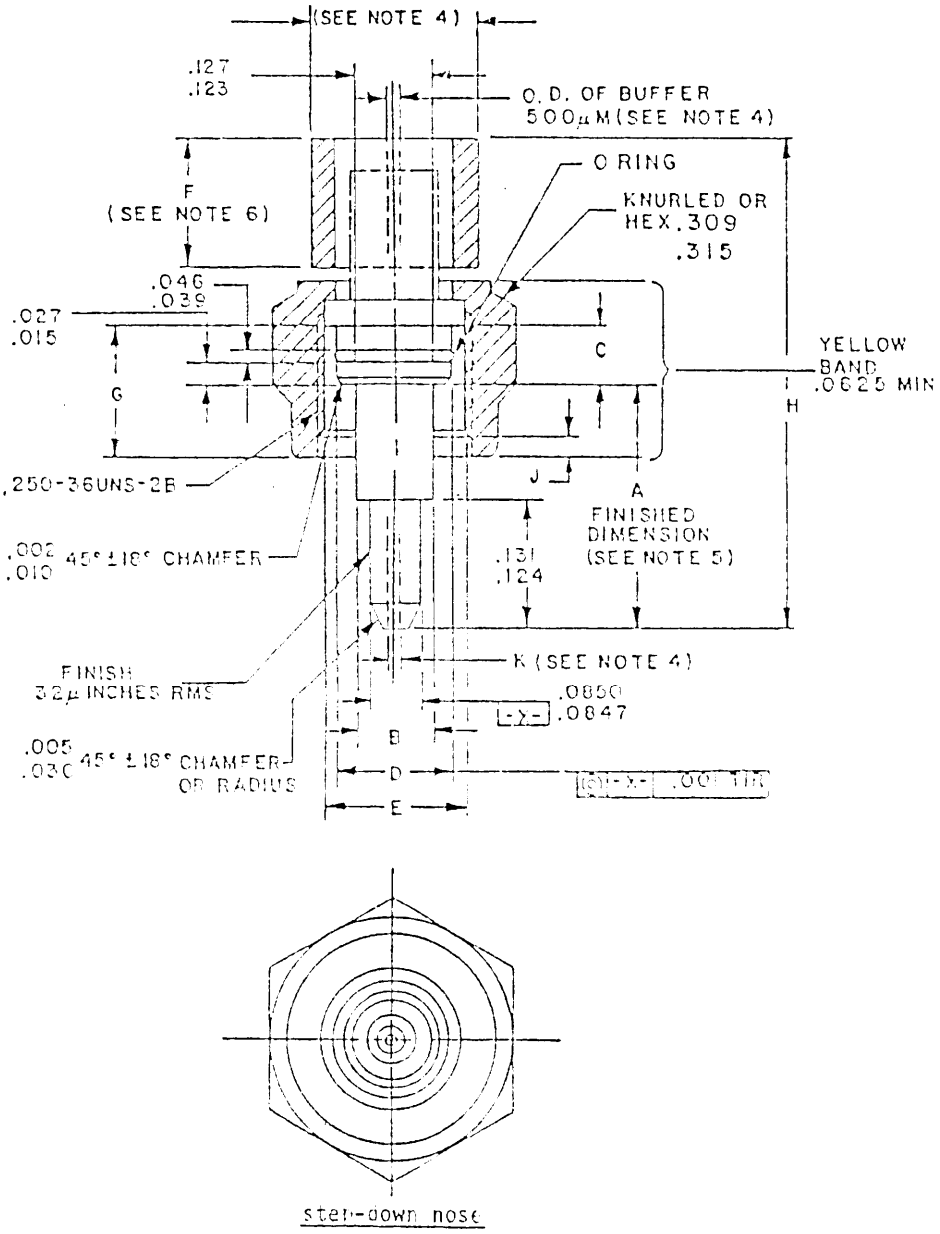


FIGURE 3. Dimensions and configuration.

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e. Safety glasses shall be worn when handling bare fibers.

	Min	Max
A	.3858 (9.804)	.3863 (9.812)
B	Plug must not enter .118 hole gauge	Plug must enter .124 Dia. hole gauge
C	.080 (2.03)	.100 (2.543)
D	.179 (4.57)	.181 (4.62)
E	.220 (5.59)	.233 (5.74)
F	.485 (12.3)	
G	.183 (4.661)	.224 (5.700)
H		1.50 (38.1)
J	.015 (0.38)	.045 (1.14)

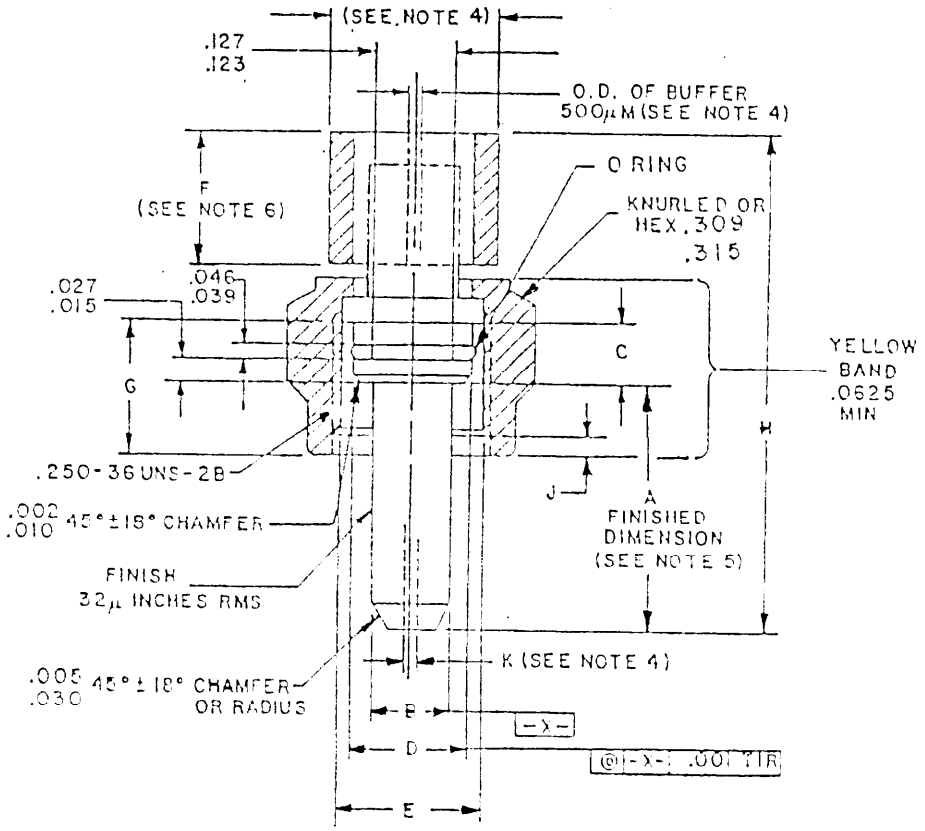
K dimension	
50/125 µm fiber	100/140 µm fiber
.0050 +.0003 -.0000	.0057 +.0003 -.0000

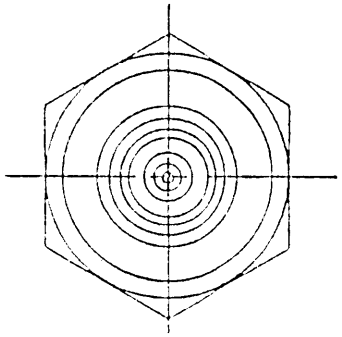
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Millimeters are in parentheses.
4. Dimensions for fiber optic cable shall be in accordance with DOD-C-85045.
5. Dimensions A is obtained after product has been assembled.
6. Ferrule may be integral part of component.
7. One full length sleeve to be used in conjunction with two step-down nose connectors.
8. One half length sleeve to be used in conjunction with one step-down nose connector and one receptacle (active device).
9. Three holes .016 (0.41 mm) minimum diameter, equally spaced, are required for safety wiring after mating. Location on the coupling nuts optional.

Dimensions and notes for step-down nose

FIGURE 3. Dimensions and configuration - Continued.

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straight nose

FIGURE 3. Dimensions and configuration - Continued.

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	Min	Max
A	.3858 (9.804)	.3862 (9.812)
B	Plug must not enter .118 hole gauge	Plug must enter .125 Dia. hole gauge
C	.080 (2.03)	.100 (2.543)
D	.179 (4.57)	.181 (4.62)
E	.220 (5.59)	.233 (5.74)
F	.485 (12.3)	
G	.183 (4.667)	.224 (5.700)
H		1.50 (38.1)
J	.015 (0.38)	.045 (1.14)

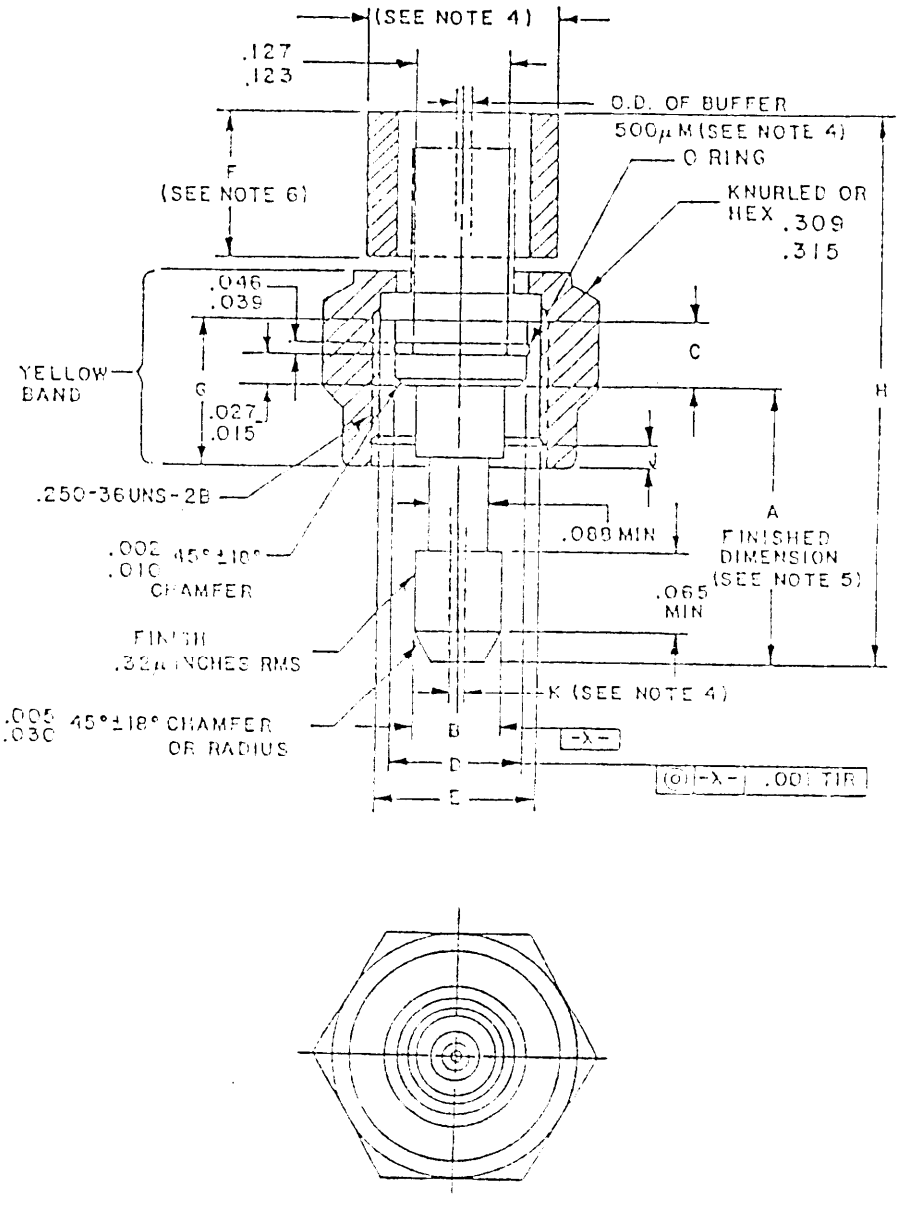
K dimension	
50/125 µm fiber	100/140 µm fiber
.0050 (0.127)	.0057 (0.145)
+.0003 (0.008)	+.0003 (0.008)
-.0000 (0.000)	-.0000 (0.000)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Millimeters are in parentheses.
4. Dimensions for fiber optic cable shall be in accordance with DOD-C-85045.
5. Dimension A is obtained after product has been assembled.
6. Ferrule may be integral part of component.
7. Three holes .016 (0.41 mm) minimum diameter, equally spaced, are required for safety wiring after mating. Location on the coupling nuts optional.

Dimensions and notes for straight nose.

FIGURE 3. Dimensions and configuration - Continued.

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Straight nose (optional view)

FIGURE 3. Dimensions and configuration - Continued.

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	Min	Max
A	.3858 (9.804)	.3863 (9.812)
B	Plug must not enter .118 hole gauge	Plug must enter .125 Dia. hole gauge
C	.080 (2.03)	.100 (2.543)
D	.179 (4.57)	.181 (4.62)
E	.220 (5.59)	.233 (5.74)
F	.485 (12.3)	
G	.183 (4.661)	.224 (5.700)
H		1.50 (38.1)
J	.015 (0.38)	.045 (1.14)

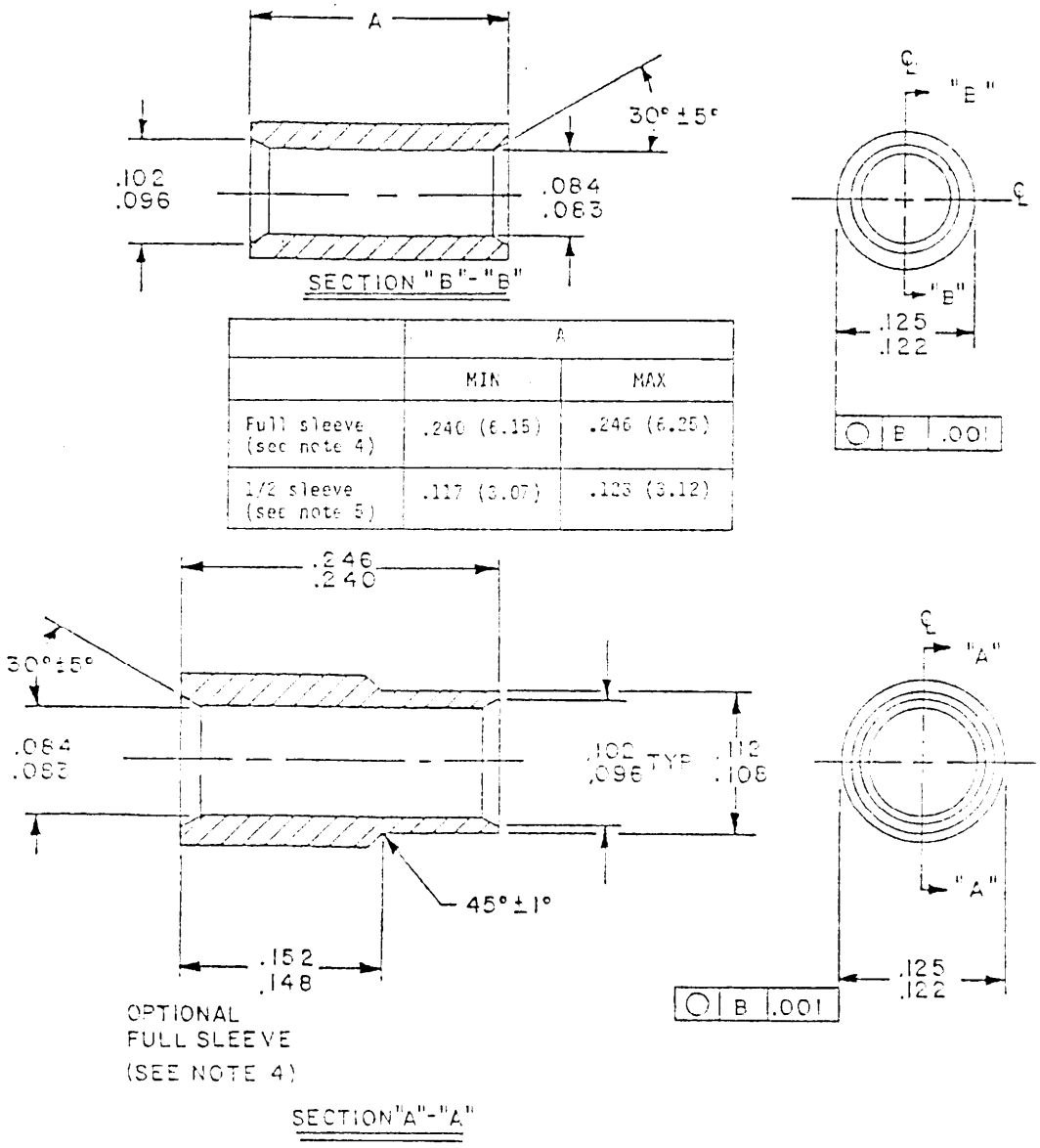
K dimension	
50/125 µm fiber	100/140 µm fiber
.0050 (0.127)	.0057 (0.145)
+.0003 (0.008)	+.0003 (0.008)
-.0000 (0.000)	-.0000 (0.000)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Millimeters are in parentheses.
4. Dimensions for fiber optic cable shall be in accordance with DOD-C-85045.
5. Dimension A is obtained after product has been assembled.
6. Ferrule may be integral part of component.
7. Three holes .016 (0.41 mm) minimum diameter, equally spaced, are required for safety wiring after mating. Location on the coupling nuts optional.

Dimensions and notes for straight nose (optional view)

FIGURE 3. Dimensions and configuration - Continued.

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	A	
	MIN	MAX
Full sleeve (see note 4)	.240 (6.15)	.246 (6.25)
1/2 sleeve (see note 5)	.117 (3.07)	.123 (3.12)

FIGURE 3A. Alignment sleeves.

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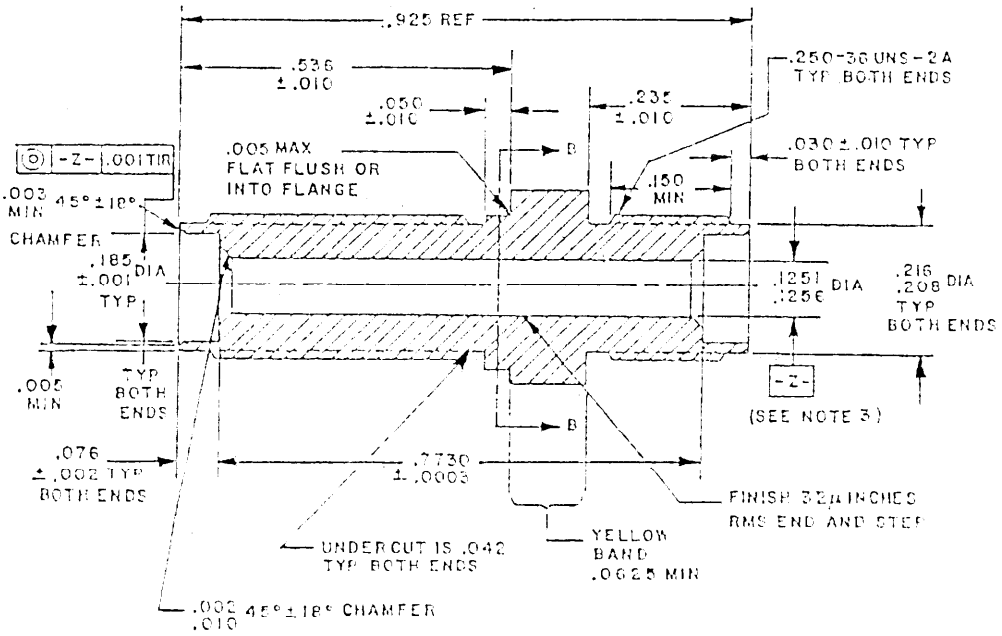
Inches	mm
.001	0.03
.083	2.11
.084	2.13
.096	2.44
.102	2.59
.108	2.74
.112	2.85
.122	3.10
.125	3.18
.148	3.76
.152	3.86
.240	6.10
.246	6.25

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents are given for general information only.
4. One full length delrin sleeve to be used in conjunction with two step-down nose connectors.
5. One half length delrin sleeve to be used in conjunction with one step-down nose connector and one receptacle (active device).
6. Dimensions apply to sleeve as manufactured. Sleeve material may be compressed when in use.

FIGURE 3A. Alignment sleeves - Continued.

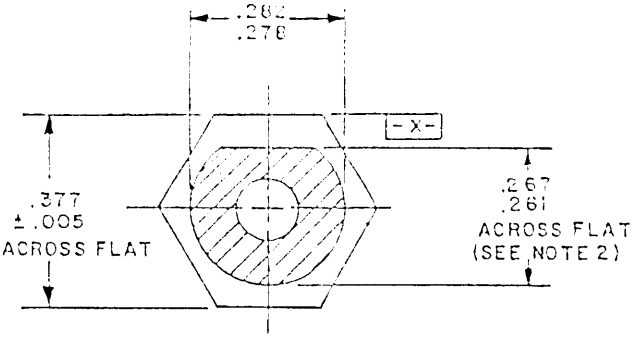
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ve sleeve. Repeat procedure on other half of sleev



SECTION A-A

Inches	mm
.0003	0.008
.001	0.03
.002	0.05
.003	0.08
.005	0.13
.010	0.25
.030	0.76
.042	1.007
.050	1.27
.0625	1.59
.076	1.93
.1251	3.178
.1256	3.190
.150	3.81
.185	4.70
.208	5.28
.216	5.49
.235	5.97
.250	6.35
.267	6.78
.278	7.06
.282	7.16
.377	9.58
.7730	19.634
.925	23.50

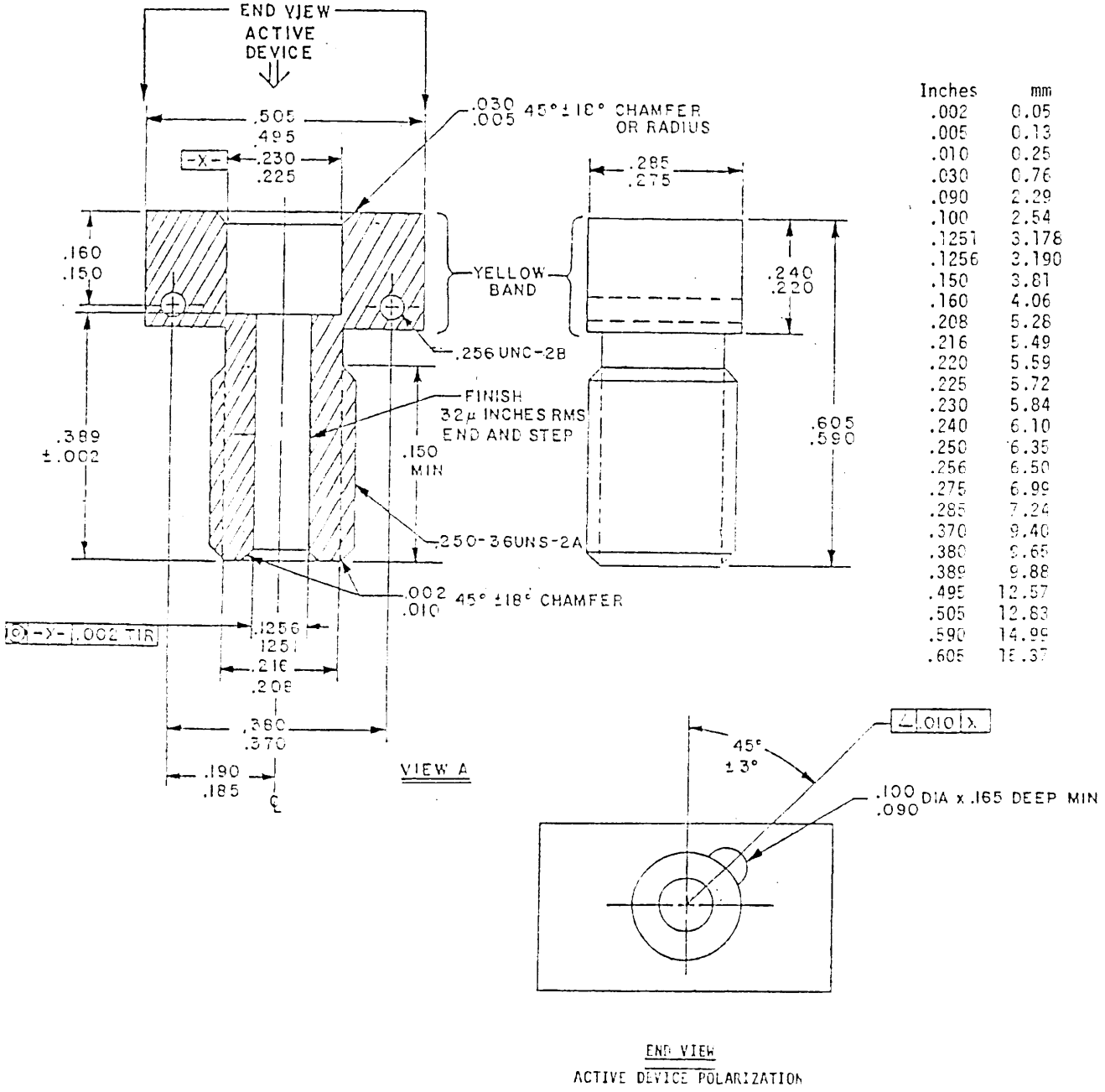


SECTION B-B

NOTES:
1. All dimensions are after plating.
2. Flat to be within ±7° of surface -X-.
3. No burrs or nicks for entire length.
4. Dimensions are in inches.
5. Metric equivalents are given for general information only.
6. Three holes .016 (0.41 mm) minimum diameter, equally spaced, are required for safety wiring after mating. Location on the coupling nuts optional.

FIGURE 4. Interface dimensions for fiber optic mating adapter, type II, class A, series 2.

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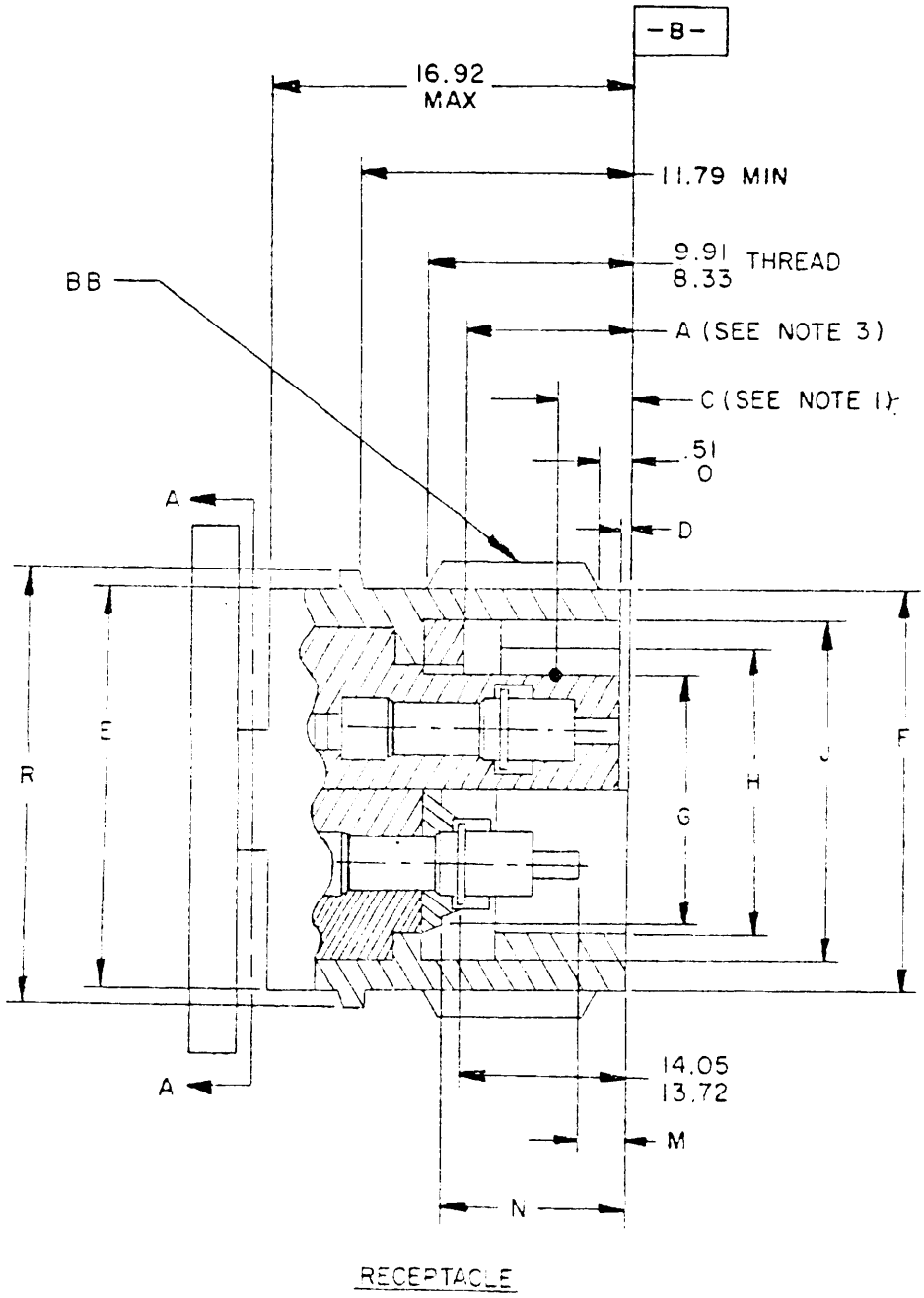


Inches	mm
.002	0.05
.005	0.13
.010	0.25
.030	0.76
.090	2.29
.100	2.54
.1251	3.178
.1256	3.190
.150	3.81
.160	4.06
.208	5.28
.216	5.49
.220	5.59
.225	5.72
.230	5.84
.240	6.10
.250	6.35
.256	6.50
.275	6.99
.285	7.24
.370	9.40
.380	9.65
.389	9.88
.495	12.57
.505	12.83
.590	14.99
.605	15.37

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 5. Interface dimensions for fiber optic connector receptacle (active device), type II, class A, series 2.

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RECEPTACLE

FIGURE 6. Intermateability dimensions for fiber optic interconnect, series 3 - type I, class A and type III, class A, style R.

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